| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2014 | **Task Last Updated:** | FY 10/08/2013 |
| **PI Name:** | Shay, Jerry W. Ph.D. | | |
| **Project Title:** | A Novel Biological Countermeasure and Mitigator of High LET-induced Cancer Progression | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | jerry.shay@utsouthwestern.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 214-648-3282 |
| **Organization Name:** | University of Texas Southwestern Medical Center | | |
| **PI Address 1:** | Cell Biology Department | | |
| **PI Address 2:** | 5323 Harry Hines Blvd | | |
| **PI Web Page:** | | | |
| **City:** | Dallas | **State:** | TX |
| **Zip Code:** | 75390-7208 | **Congressional District:** | 30 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N |
| **Start Date:** | 01/01/2011 | **End Date:** | 12/31/2014 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | 4 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Simonsen, Lisa | **Contact Phone:** | |
| **Contact Email:** | lisa.c.simonsen@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date is now 12/31/2014 per NSSC information (Ed., 1/22/14) | | |
| **Key Personnel Changes/Previous PI:** | Ugur Eskiocak was a doctoral student who has completed his PhD, Sept 2011. He just left the lab to begin a postdoctoral fellowship in the laboratory of Sean Morrison. Other students are involved in specific aspects of this project (Peter Ly, completed his PhD, Nov 2012 and is currently in Don Cleveland's lab at the Ludwig Institute for Cancer Research at the University of California, San Diego, and Lu Zhang in her 4th years as a graduate student in the lab). Together with Aadil Kaisani (another graduate student, completing his PhD May 2014) and the addition of Mariam El-Ashmawy (MD/PhD graduate student), we have a solid team of students and postdoctoral fellows (Dr. Sang Kim) assisting in this project. | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX11AC15G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | This proposal addresses Radiation Carcinogenesis Cancer Specific Gap 8: Biological Countermeasures (BCM). While progress has been made since NCRP report 132 in 2000, we still do not have the data for assessing the need for BCMs. However, particle emission levels for fatal cancer risks are projected to be violated for a Mars mission under all possible scenarios. With current permissive exposure limits most astronauts would be limited to about 6-8 months on deep space missions. Thus, if a goal of NASA is to have experienced crews fly multiple times, it is prudent to assume that even with new research to modify uncertainty reductions in fatal cancer risk projection models, that the development of BCMs should now be initiated.<br>The work in progress has been to investigate Bardoxylone methyl (BARD), an oral available non toxic anti-oxidant inflammatory modulator that is currently in Phase IIB clinical trials for patients with renal insufficiency. BARD is a synthetic triterpenoid and there is a significant molecular understanding of its mode of action. We have already completed investigations with low and high LET irradiation in both normal human lung and colonic epithelial cells and have shown that 10-50nM BARD is an effective radiation BCM, reducing double-strand DNA damage and anchorage-independent cancer progression. Based on our results in human epithelial cells, we now propose to demonstrate BARD is effective in mouse models of cancer. We have a commitment from Reata Pharmaceuticals (Irving, TX) to provide the compound required for animal testing. We believe BARD has an excellent chance of being developed into a potent BCM to protect against high LET irradiation. There are data showing that BARD can protect against radiation-induced mucositis in a hamster cheek pouch assay. In addition, BARD protects against radiation-induced intestinal damage in zebrafish development. Lastly, we have evidence that BARD can mitigate gamma-irradiation (using the colony formation assay) if provided within 30 minutes after IR. In summary, the main objective of this V-NSCOR application is to investigate if BARD is a potent BCM and/or mitigator in mouse models that are predisposed to cancer.<br><br>Specific aim 1: To use a lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high LET is provided in fractionated/protracted doses over 5 to 10 days (e.g 0.2 Gy x 5 days or 0.1 Gy x 10 days 56Fe 1GeV, compared to single acute 1 Gy 56Fe). Our objective is to determine if pretreatment with BARD prior to acute or fractionated IR reduces the number of invasive (e.g. more lethal) cancers. We will examine at least 3 different ions.<br><br>Specific aim 2: To test BARD as a BCM in mouse models exposed to SPE simulations. As part of the UT Southwestern Lung NSCOR and the Georgetown GI NSCOR, we have lung cancer and colon cancer susceptible mice that have been treated with solar particle simulations. We propose to test if providing BARD to these animals before irradiation reduces the incidence or progression of cancer.<br><br>Specific aim 3: To test BARD as a radiation mitigator by providing it to mice within 60 minutes after irradiation. We will compare SPE simulations (total body exposure 2 Gy Protons) and at least two other ions (56Fe at 600 meV, LET 175 and 28Si at 300 MeV LET 70).<br><br>The significance of the proposed work is that we will be able to demonstrate if BARD reduces cancer incidence or progression in two important tissues (lung and colon) that are susceptible to radiation-associated cancer using doses and doses rates recommended by NASA. This research should provide new data to help reduce uncertainties in current risk projection models. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This project may lead to the development of a non toxic, oral available radiation countermeasure and mitigator that will not only be important to protect astronauts from the detrimental effects of space radiation, but may also have utility in protecting normal human tissues when radiotherapy is used to treat cancer patients. |
| | This proposal addresses Radiation Carcinogenesis Cancer Specific Gap 8: Biological Countermeasures. While significant progress has been made since the NCRP report 132 in 2000, particle emission levels for fatal cancer risks are projected to be violated for a Mars mission under all possible scenarios, unless risk estimates are lowered considerably or the increased risk of potential fatal cancers is increased. It generally accepted that there appears to be no threshold below which radiation exposure can be viewed as harmless (BEIR VII, 2005). With current permissive exposure limits, most astronauts would only be able to remain for about 6-8 months on deep space missions. Thus, if a goal of NASA is to have crews fly multiple times and on long missions, it is prudent to assume, even with new research to modify uncertainty reductions in fatal cancer risk projection models, that it is appropriate to conduct research into the development of space radiation biological countermeasures. Our work has been to investigate a family of plant titerpenoids (originally derived from Spanish moss) that have been synthesized and studied for radiation protection and mitigation experiments. Specifically we have conducted low and high LET IR experiments in human lung and colonic epithelial cell lines and in mouse models of lung and colon cancer susceptibility. Our cell culture studies during the first year of this project involved the use of bardoxolone methyl (BARD, also called CDDO-Me) and an analogue (CDDO-EA), both oral available non-toxic anti-oxidant inflammatory modulators (AIMs). BARD is a synthetic triterpenoid and there is a significant molecular understanding of its mode of action. We have already completed a series of investigations with low-LET and high-LET irradiation in both normal human lung and colonic epithelial cells and have published that single doses of 10-50nM BARD/CDDO-Me are effective radiation biological countermeasures, reducing double-strand DNA damage and anchorage-independent cancer progression. In mice we have also shown CDDO-EA may also have a mitigation effect by enhancing repair of DNA damage.<br>Criteria for approval by the FDA for BARD as a biological countermeasure can be met based on current data except for protection from high-LET irradiation in an animal model. We have conducted experiments and now have robust results indicating that CDDO-EA partially protects induced invasive cancer from proton irradiation both single doses and provided as a simulated solar particle event. We also have experimental evidence that BARD can mitigate steps in cancer progression from gamma-irradiation (using colony formation assays) if provided 0.5-12 hr after IR via a novel mechanism of action that involve enhanced DNA damage repair and this work was published during the last review period. In summary, the main objective of this project is to investigate if CDDO-EA is a potent biological countermeasure and/or mitigator of SPE simulations and NASA priority HZE ions in mouse models that are predisposed to solid cancers. Since BARD is an anti-inflammatory modulator of oxidative damage, we believe BARD may also protect against HZE induced short-term and long-term neurodegenerative cognitive deficits and perhaps cardiovascular injury. We have already conducted a significant amount of research at the NSRL that support approaches to moving CDDO-EA forward as a lead space radiation biological countermeasure candidate. Our hypothesis is that |

| | |
|---|---|
| **Task Progress:** | BARD/CDDO-EA may lower the HZE carcinogenic risks. Our proposed research is hypothesis based, focuses on solid cancer risks, and is mechanistically driven. W now need to conduct lower dose experiments with high LET IR using additional NASA high priority ions. The specific aims are:<br><br>Specific Aim 1: To use the LA-1 lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high-LET is provided in fractionated/protracted doses. Our objective is to determine if pretreatment with non toxic doses of CDDO-EA prior to acute or fractionated HZE irradiation reduces the number of invasive (e.g. more lethal) lung cancers. We will compare this to other priority HZE ions using fluences and energies that are representative of GCR including 28Si and 16O. We will also test a mouse model of colon cancer (CPC;APC).<br><br>Specific aim 2: To test CDDO-EA as a biological countermeasure in mouse models exposed to SPE simulations.<br><br>Specific aim 3: To test CDDO-EA as a radiation mitigator by providing it to mice within 30-60 minutes after irradiation (via IP and orally administered gavage). We will compare SPE simulations (total body exposure 2 Gy protons) and two other ions (56Fe at 600 MeV/n, LET 175 at dose ranges between 0.1 Gy and 0.5 Gy and 28Si at 350-600 MeV/n LET 70 at doses between 0.2 and 0.8 Gy). We will test single acute versus fractionated doses of 56Fe and 28Si, having the expected outcomes (even with very short-term treatments).<br><br>The summary of progress in the lung cancer mouse model:<br><br>• 2Gy SPE and 1 Gy fractionated Fe- irradiation decreases the lifespan of LA1 K-ras animals and results in an increase incidence of carcinoma.<br><br>• LA1 K-ras mice on CDDO-EA diet during fractionated Fe- irradiation exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet.<br><br>• LA1 K-ras mice on CDDO-EA diet during solar particle irradiation simulations exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet.<br><br>• LA1 K-ras mice fed CDDO-EA without irradiation show reduced tumor progression.<br><br>The summary of progress in the colon cancer mouse model:<br><br>• 2 Gy SPE - irradiation results in a 3-fold increase incidence of carcinoma.<br><br>• The CPC;APC mice on CDDO-EA diet during solar particle irradiation simulations exhibit fewer polyps ~100 days post –IR in comparison to mice on control diet.<br><br>• The CPC;APC mice on CDDO-EA diet during solar particle irradiation simulations exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet ~250 days post-IR. |
| **Bibliography Type:** | Description: (Last Updated: 11/21/2025) |
| **Abstracts for Journals and Proceedings** | Kaisani A, Kim SB, Delgado O, Fasciani ODG, Batten K, Richardson JA, Wright WE, Story MD, Fornace AJ, Minna JD, Shay JD. "Risk Assessment of Radiation Induced Invasive Cancer in Mouse Models of Lung and Colon Cancer." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. Proceedings of the symposium distributed. , May-2013 |
| **Articles in Peer-reviewed Journals** | Hunt CR, Ramnarain D, Horikoshi N, Iyengar P, Pandita RK, Shay JW, Pandita TK. "Histone modifications and DNA double-strand break repair after exposure to ionizing radiations." Radiat Res. 2013 Apr;179(4):383-92. Epub 2013 Feb 1. Review. http://dx.doi.org/10.1667/RR3308.2 ; PubMed PMID: 23373901 , Apr-2013 |
| **Articles in Peer-reviewed Journals** | Shay JW. "Determining if telomeres matter in colon cancer initiation or progression." J Natl Cancer Inst. 2013 Aug 21;105(16):1166-8. http://dx.doi.org/10.1093/jnci/djt200 ; PMID: 23918446 , Aug-2013 |
| **Articles in Peer-reviewed Journals** | Kim SB, Ly P, Kaisani A, Zhang L, Wright WE, Shay JW. "Mitigation of radiation-induced damage by targeting EGFR in non-cancerous human epithelial cells." Radiation Research. 2013 Sep;180(3):259-67. http://dx.doi.org/10.1667/RR3371.1 ; PubMed PMID: 23919312 , Sep-2013 |
| **Articles in Peer-reviewed Journals** | Kim SB, Pandita RK, Eskiocak U, Ly P, Kaisani A, Kumar R, Cornelius C, Wright WE, Pandita TK, Shay JW. "Targeting of Nrf2 induces DNA damage signaling and protects colonic epithelial cells from ionizing radiation." Proc Natl Acad Sci U S A. 2012 Oct 23;109(43):E2949-55. Epub 2012 Oct 8. http://dx.doi.org/10.1073/pnas.1207718109 ; PMID: 23045680 . , Oct-2012 |